SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:

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ASSESSMENT TOOL 1

COURSE NAME: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA-1709

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ANALYTICAL PROBLEM SOLVING

QUESTION 1: WATER JUG PROBLEM

The objective is to obtain exactly 2 gallons in the 4-gallon jug using a 4-gallon jug and a 3-gallon jug. The jugs have no measuring marks, and the permitted operations are filling a jug from the pump, emptying water onto the ground, and pouring water from one jug into the other. Represent each state as (x, y), where x is the amount in the 4-gallon jug and y is the amount in the 3-gallon jug.

Initially, both jugs are empty, so the state is (0, 0). Step 1: Fill the 3-gallon jug, producing (0, 3). Step 2: Pour the 3-gallon jug into the 4-gallon jug, producing (3, 0). Step 3: Fill the 3-gallon jug again, producing (3, 3). Step 4: Pour from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full. Only 1 gallon fits, so the resulting state is (4, 2). Step 5: Empty the 4-gallon jug, producing (0, 2). Step 6: Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug, producing (2, 0). The 4-gallon jug now contains exactly 2 gallons.

Step	Operation	State (4-gal, 3-gal)	Result
0	Start	(0, 0)	Both empty
1	Fill 3-gallon jug	(0, 3)	3-gallon jug full
2	Pour 3-gallon into 4-gallon	(3, 0)	4-gallon jug has 3
3	Fill 3-gallon jug	(3, 3)	Both contain 3
4	Pour into 4-gallon until full	(4, 2)	2 gallons remain in 3-gallon jug
5	Empty 4-gallon jug	(0, 2)	2 gallons remain in 3-gallon jug
6	Pour 3-gallon into 4-gallon	(2, 0)	Goal achieved

Final answer: The state (2, 0) means the 4-gallon jug contains exactly 2 gallons and the 3-gallon jug is empty.

QUESTION 2: MARS ROVER AS AN INTELLIGENT AGENT

A Mars Rover is an autonomous robotic agent that senses its surroundings, moves across the Martian surface, examines rocks and soil, and collects or analyzes samples. Its decisions must account for limited communication with Earth, uncertain terrain, energy limits, scientific priorities and equipment safety.

(i) Percepts of the agent

The rover's percepts include camera images, terrain and obstacle measurements, wheel slip, tilt and orientation, distance measurements, temperature, battery and power status, instrument readings, and the chemical or mineral composition detected in rocks and soil. These percepts allow the rover to estimate its location, recognize hazards, identify scientifically interesting targets, and monitor its own condition.

(ii) Operating environment

The environment is partially observable because sensors cannot reveal every detail of the terrain or the interior composition of a sample. It is sequential because each movement affects future positions, energy use and available scientific opportunities. It is dynamic because environmental and rover conditions can change. It is continuous in physical space and time, and it is uncertain because terrain interactions and scientific measurements may not be perfectly predictable. The rover is also a multi-stage autonomous system with communication delays, so it cannot rely on immediate human instructions for every action.

(iii) Actions the agent can take

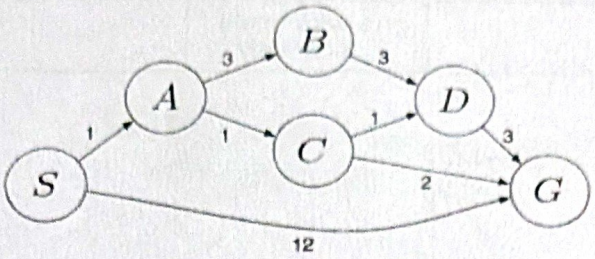
The rover can move forward or backward, turn, stop, choose a route, capture images, focus or reposition cameras, use a robotic arm, drill or abrade a rock when equipped, collect or place a sample, run onboard scientific instruments, transmit data to Earth, conserve power, and enter a safe mode when a fault or hazard is detected. The exact actions depend on the rover's hardware and mission commands.

(iv) Performance evaluation

Performance can be evaluated by the quality and quantity of useful scientific data, the number and relevance of samples analyzed, successful completion of mission objectives, accuracy of navigation, distance traveled without damage, energy efficiency, communication and data-transfer success, equipment reliability, and the ability to avoid hazards. A good performance measure rewards meaningful scientific discoveries and safe mission completion rather than movement alone.

(v) Most suitable agent architecture

A hybrid architecture combining a model-based, goal-based and utility-based agent is most suitable. A model-based component maintains an internal map and a record of rover state because the environment is only partially observable. A goal-based component plans actions toward mission objectives, such as reaching a target or analyzing a sample. A utility-based component compares alternatives using scientific value, travel risk, energy consumption, time and hardware safety. A learning component can further improve terrain classification and target selection from experience.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select metho
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculation
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretati
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterpr or unable understa problem